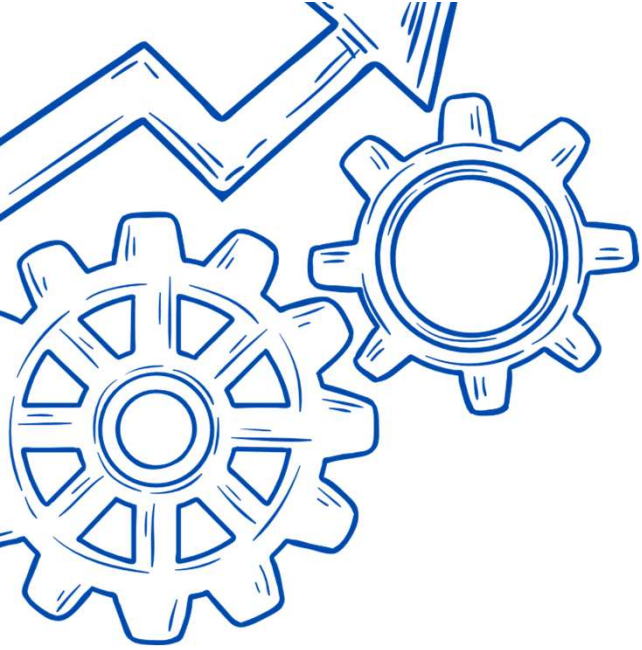
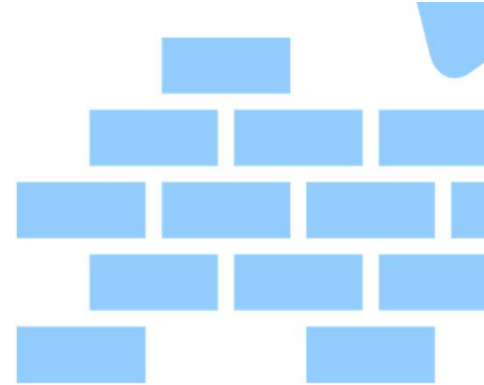


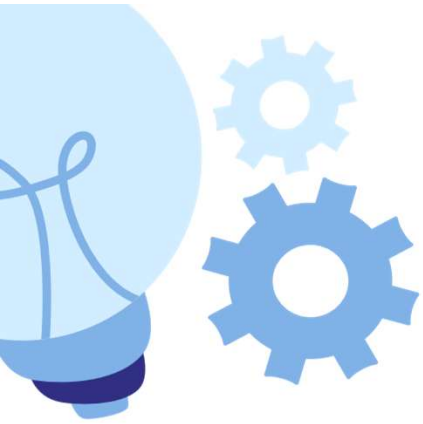


PCB Manufacturing System with Laser Engraving

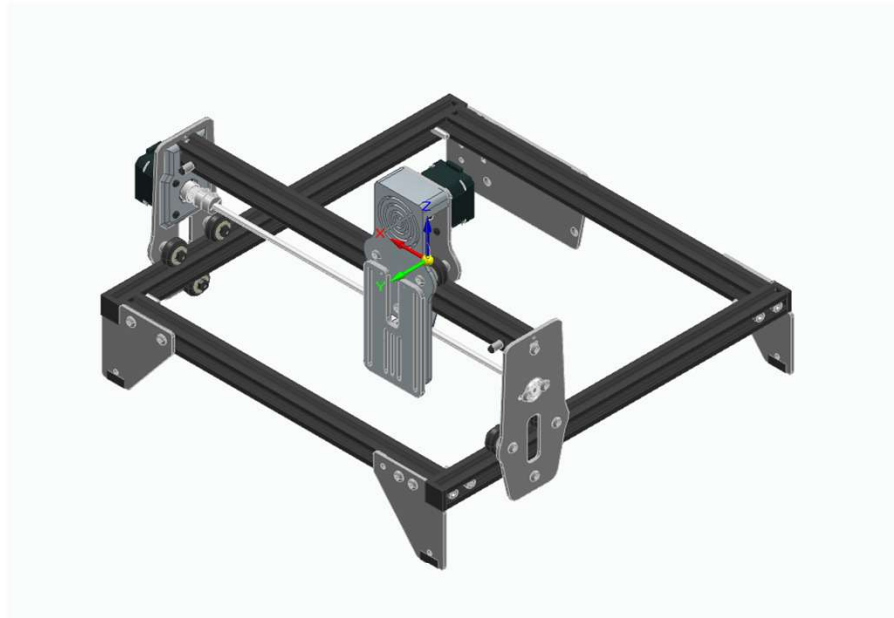
Ahmet Sadaka

Marmara University - Mechatronics Engineering

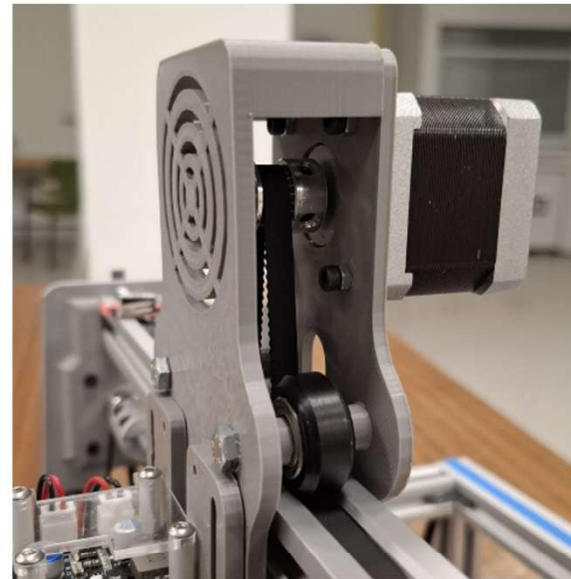


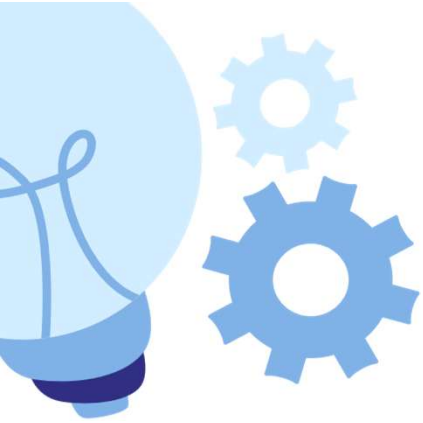


Mechanical Design and Chassis



Mechanical Design and Chassis





Electronic System and Control Unit

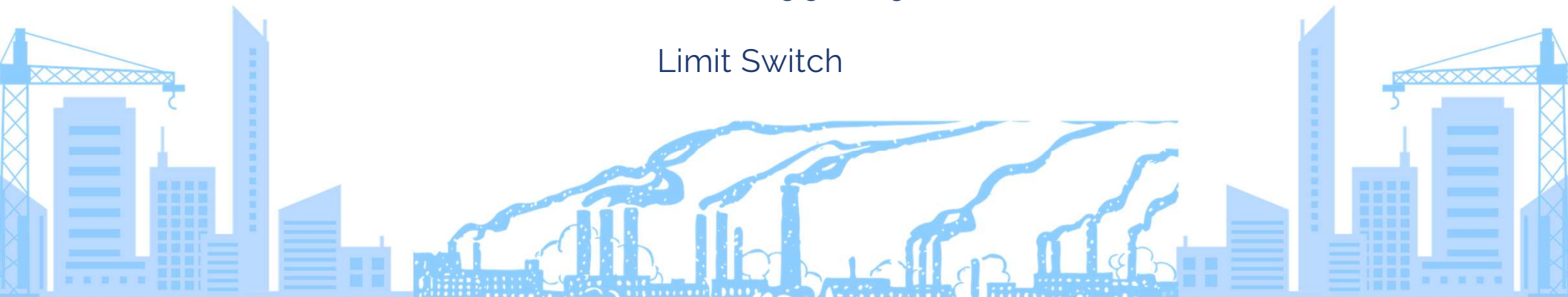
Controller: STM32F103C8T6

Motor Drivers: DRV8825 modules.

Laser: Lt-20w_a

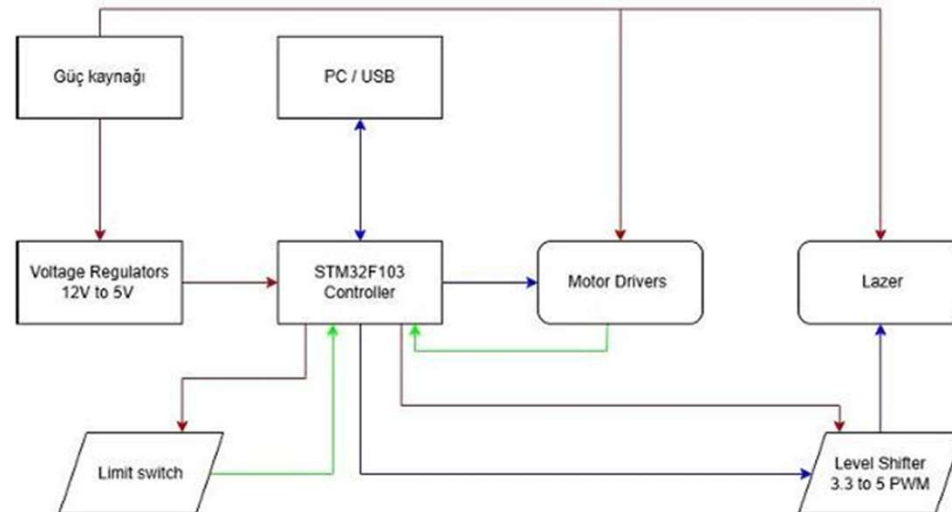
Level Shifter 3.3v $\rightarrow$ 5v

Limit Switch



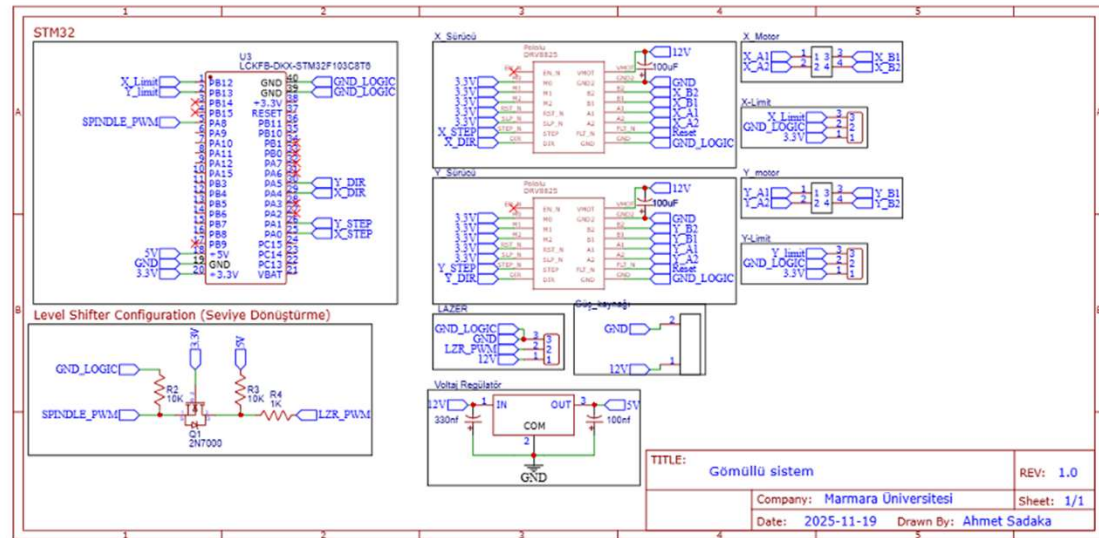
Electronic System and Control Unit

System block diagram



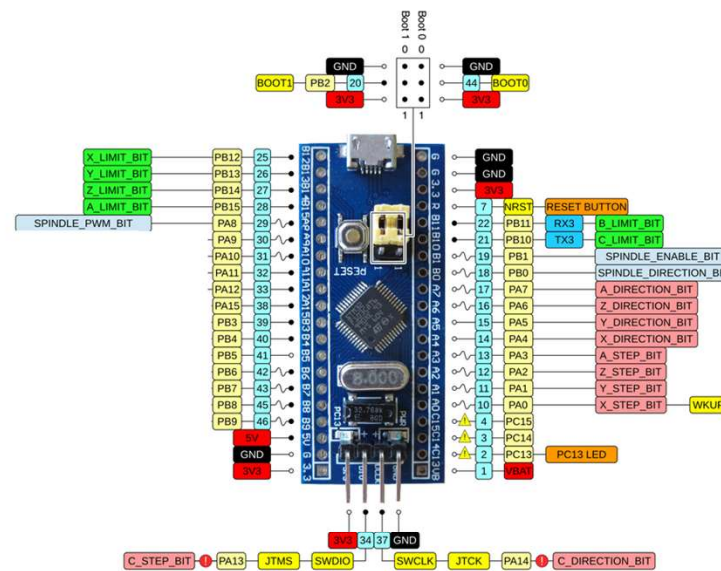
Electronic System and Control Unit

System wiring diagram



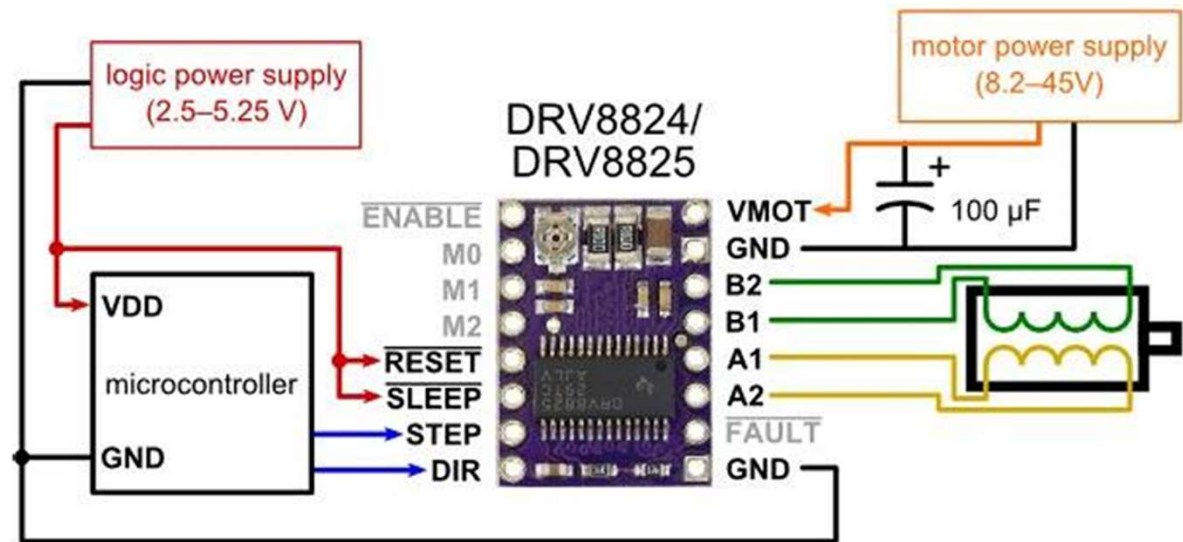
Electronic System and Control Unit

STM32 Pin Schematic



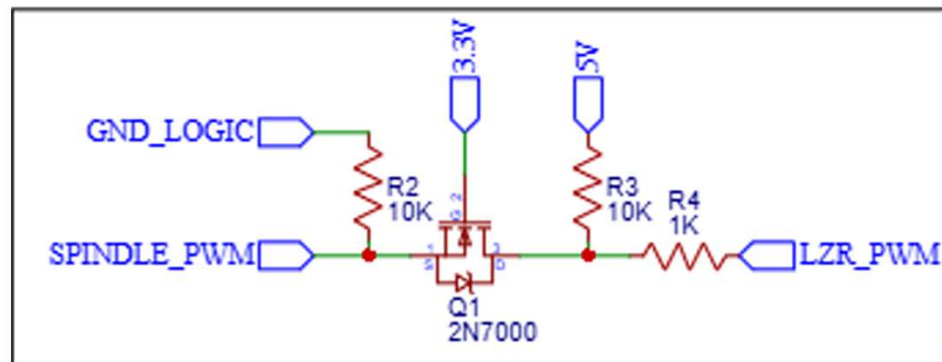
Electronic System and Control Unit

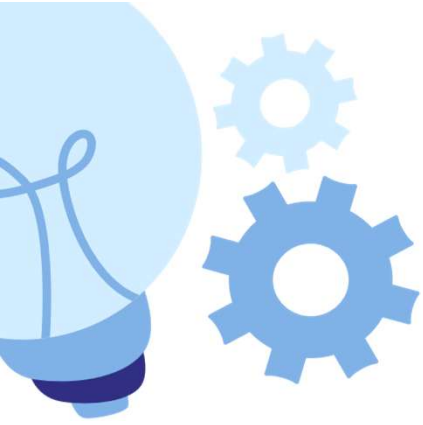
DRV8825 wiring
Diagram



Electronic System and Control Unit

Level Shifter





Software Architecture

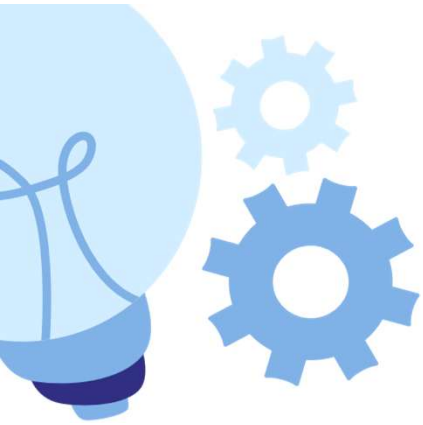
Movement Planning and
Control

G-Code Processing and
Communication

Peripheral Unit and
Laser Management

System and
Configuration





Software Architecture

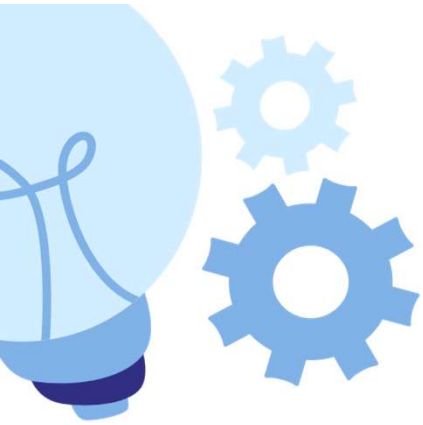
Movement Planning
and Control

Planner.c / Planner.h

Stepper.c / Stepper.h

Motion_control.c





Software Architecture

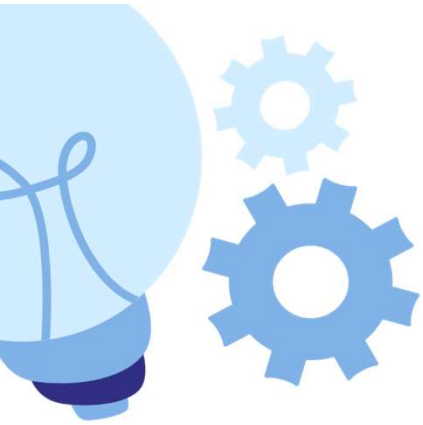
G-Code Processing
and Communication

gcode.c / gcode.h

protocol.c

serial.c:





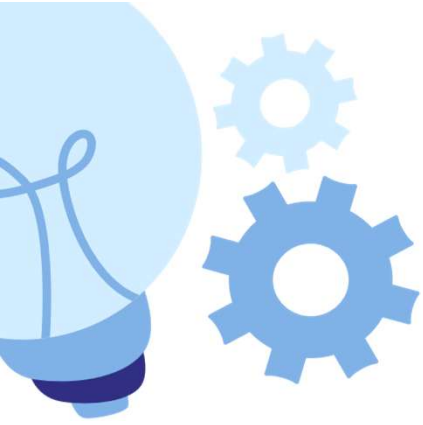
Software Architecture

Peripheral Unit and
Laser Management

`spindle_control.c:`

`limits.c:`





Software Architecture

System and
Configuration

config.h:

defaults.h

cpu_map.h

